
(a)
L
2
3
E_J
C_J
C
1
4
(b)
0
π
θ
−2π
ϕ
2π
π
θ
0
−2π
ϕ
2π
(c)
|0⟩→|0_L⟩
|2⟩→|1_L⟩
π
θ
0
−π
ϕ
π
(d)
E
−0.5
$\phi_{ext}/2\pi$
0.5
E_4
E_3
E_2
E_1
E_0
E
−0.5
n_g^θ
0.5